

The Resonance Cartographer: Mapping the Information Singularity via Topological Bridges in Cosmology and Quantum Gravity

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Abstract

We propose the "Information Singularity" hypothesis, a theoretical framework suggesting that the origin of the universe ($t \rightarrow 0$) is not a point of infinite matter density, but a state of maximum, finite information density bounded by holographic limits. By mapping a topological database of 2,772 autonomously extracted physics equations across three scales (Fundamental Physics, Advanced Frameworks, and Emergence/Intelligence), we identify mathematical variables that survive the cosmic rewind. We formulate a modified Friedmann equation coupling Integrated Information (Φ) to the Einstein Field Equations via a Landauer-bound stress-energy tensor. We suggest that quantum gravity frameworks mathematically enforce a Holographic Ceiling, preventing the singularity. Finally, we explore a thermodynamic bridge showing that the dissipative entropy production of the human brain and the Bekenstein-Hawking entropy of the cosmic web share structural mathematical analogies, suggesting the possibility of a unified computational substrate for reality.

1 Introduction

The standard cosmological model posits that the universe began in a state of infinite density and temperature—a gravitational singularity. However, the mathematical breakdown of General Relativity at the Planck scale ($t \rightarrow 0$) suggests that classical concepts of matter and energy are insufficient to describe the origin.

In this paper, we explore the hypothesis that the universe is fundamentally an information-processing system, and that its origin is an *Information Singularity*. We leverage a topological database of 2,772 equations extracted by autonomous AI agents to identify mathematical resonances across disparate fields of physics. Here, the database relationships were mapped using graph-theoretic topology, where nodes represent physical variables and edges represent functional dependencies. Mathematical resonances were identified by searching for graph isomorphic substructures, revealing recurring scaling behaviors across disparate scales. Using this method, we find that Integrated Information (Φ) [10], typically used to measure consciousness in neural systems, shares a structural mathematical analogy with the scalar fields driving cosmic inflation [9].

By rewinding a modified Friedmann equation to $t \rightarrow 0$, we show that classical matter vanishes, leaving Information Density (ρ_Φ) as the sole driver of the Big Bang. We then utilize thermodynamic data to bridge the entropy production rates of biological neural networks to the macroscopic entropy of the cosmic web, providing a multi-scale validation of the universe as a computational engine.

2 The Information Tensor and the Singularity Rewind

To formalize the Information Singularity, we introduce a modified Einstein Field Equation coupling spacetime geometry to an information stress-energy tensor, $T_{\mu\nu}^\Phi$:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G (T_{\mu\nu}^{\text{matter}} + \lambda T_{\mu\nu}^\Phi) \quad (1)$$

By the Bianchi identity ($\nabla^\mu G_{\mu\nu} = 0$), the total energy-momentum tensor must be conserved: $\nabla^\mu (T_{\mu\nu}^{\text{matter}} + \lambda T_{\mu\nu}^\Phi) = 0$. Assuming no direct energy-momentum exchange between the matter and information sectors (analogous to non-interacting dark matter and dark energy), each is independently conserved ($\nabla^\mu T_{\mu\nu}^\Phi = 0$).

To motivate the form of $T_{\mu\nu}^\Phi$ from an action principle, we represent the information sector by a minimally coupled scalar field $\Phi(x)$ governed by the action $S_\Phi = \int d^4x \sqrt{-g} \mathcal{L}_\Phi$, where the Lagrangian density is:

$$\mathcal{L}_\Phi = -\frac{1}{2} g^{\alpha\beta} \nabla_\alpha \Phi \nabla_\beta \Phi - V(\Phi) \quad (2)$$

Varying this action with respect to the metric yields the information stress-energy tensor:

$$T_{\mu\nu}^\Phi = \nabla_\mu \Phi \nabla_\nu \Phi - g_{\mu\nu} \left(\frac{1}{2} g^{\alpha\beta} \nabla_\alpha \Phi \nabla_\beta \Phi + V(\Phi) \right) \quad (3)$$

In the cosmological background, this reduces to an equivalent perfect fluid with density $\rho_\Phi = \frac{1}{2} \dot{\Phi}^2 + V(\Phi)$ and pressure $p_\Phi = \frac{1}{2} \dot{\Phi}^2 - V(\Phi)$. Here, the coupling constant λ has dimensions of energy per unit of information (Joules/bit). In natural units ($c = \hbar = k_B = 1$), λ is a dimensionless scaling parameter of order unity, setting the scale where information density couples to spacetime curvature with equal weight to baryonic matter.

Rewinding to the origin state ($t \rightarrow 0$), classical matter density goes to zero ($T_{\mu\nu}^{\text{matter}} \rightarrow 0$). The expansion rate H^2 is driven entirely by the Information Density ρ_Φ . Using Landauer's Principle ($E = k_B T \ln 2$), we translate this into a literal Bit-Density. Throughout this paper, Φ denotes a formal information-theoretic measure representing the total integrated information content of a given spacetime region. While inspired by Tononi's Integrated Information Theory [10], we do not claim that the early universe possesses consciousness; rather, Φ serves as a mathematical variable capturing the degree of causal structure and information integration in the cosmic medium. To prevent the thermodynamic temperature T from diverging classically ($T \rightarrow \infty$), we assume quantum gravity effects cap the temperature at the Planck temperature ($T_{\text{Pl}} \approx 1.4 \times 10^{32}$ K, consistent with cosmological parameters [9]), which serves as a natural ultraviolet thermal cutoff. The event horizon integrated information Φ represents the total holographic degrees of freedom of the boundary, consistent with physical bounds on the computational capacity of the universe [17]. The energy density is then:

$$\rho_\Phi = \frac{\Phi \cdot k_B T_{\text{Pl}} \ln 2}{V} = \frac{3H^2}{8\pi G} = \rho_c \quad (4)$$

Rather than being a circular parameter mapping, this equivalence arises naturally from the holographic principle [13, 14], which bounds the maximum information density of a Planck-scale volume to the Planck density ($\rho_{\text{Pl}} \sim \rho_c$ at the Planck epoch). At the origin, Information Density exactly saturates the Critical Density (ρ_c), suggesting that an information-dominated state can naturally source the initial cosmological expansion.

3 Comparative Thermodynamics: From Neural Networks to the Cosmic Web

To validate the information-theoretic nature of reality, we analyze a mathematical bridge regarding entropy production rates in neural information processing versus gravitational structure formation.

3.1 Neural Information Processing

The adult human brain consumes roughly $P_{\text{brain}} \sim 20 \text{ W}$ of total metabolic power [1]. While Attwell & Laughlin estimate signaling costs at $\sim 1 \text{ W}$, the total thermodynamic dissipation sets the absolute upper bound for entropy production. If most metabolic power ultimately becomes heat at body temperature ($T \approx 310 \text{ K}$), the thermodynamic entropy production rate is:

$$\dot{S}_{\text{brain}} \approx \frac{P_{\text{diss}}}{T} \approx \frac{20 \text{ J s}^{-1}}{310 \text{ K}} \approx 6.5 \times 10^{-2} \text{ J K}^{-1} \text{ s}^{-1} \quad (5)$$

In terms of Boltzmann's constant, this is $\dot{S}_{\text{brain}} \sim 4.7 \times 10^{21} k_B \text{ s}^{-1}$. The minimum thermodynamic cost of erasing one bit of information is bounded by Landauer's principle [2]:

$$W_{\text{min}} = k_B T \ln 2 \quad (6)$$

At $T = 310 \text{ K}$, this yields $k_B T \ln 2 \approx 2.9 \times 10^{-21} \text{ J/bit}$. Biological neural systems operate far above this limit, maintaining nonequilibrium steady states through continuous ATP consumption. This nonequilibrium behavior is characterized by a broken detailed balance, as demonstrated in both human neural time-series [3] and balanced spiking networks [4].

3.2 Gravitational Structure Formation and Black-Hole Entropy

In contrast, the mainstream Λ CDM cosmological model describes structure formation via gravitational instability. While collisionless dark matter conserves fine-grained Gibbs entropy, coarse-grained entropy increases through phase mixing and violent relaxation. Alternative formulations for gravitational entropy include Penrose's Weyl curvature hypothesis [7], which posits that the initial singularity has vanishing Weyl curvature ($C_{\mu\nu\rho\sigma} \rightarrow 0$), and that entropy grows as matter clumping increases anisotropy. Alternatively, Hosoya, Buchert, and Ehlers [8] define gravitational entropy through the spatial volume-averaging of cosmological backreaction in inhomogeneous spacetimes. The cleanest gravitational entropy is black-hole entropy [6], which fundamentally relies on the Holographic Principle [13, 14]:

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar} \quad (7)$$

Egan and Lineweaver [5] estimated that the entropy of the observable universe is dominated by supermassive black holes, with a total entropy of order:

$$S_{\text{universe}} \sim 10^{104} k_B \quad (8)$$

3.3 The Thermodynamic Bridge and Its Limits

The comparative analysis reveals a profound symmetry. Neural systems create local order by exporting entropy as heat, strictly bounded by Landauer's limit. The cosmic web creates gravitational structure while increasing coarse-grained, radiative, or black-hole entropy. However, it is crucial to note the limits of this analogy: while biological brains are coupled to a localized 310 K heat bath, cosmic structures decohere through interactions with the expanding spacetime background. Despite these physical differences, both systems represent self-organizing networks whose dynamics can be modeled via open-system Lindblad master equations. This connection (seen in spiking networks [4] and cosmological dephasing) should therefore be viewed as a structural mathematical analogy rather than a direct physical equivalence.

4 The Quantum Hardware: Enforcing the Holographic Ceiling

If the universe begins as a finite grid of quantum information saturating the critical density (ρ_c), classical general relativity must be modified at the Planck scale to prevent infinite singularities. We propose that this "Holographic Ceiling" is mathematically enforced by the convergence of two quantum gravity frameworks: Non-Commutative Geometry (NCG) [11] and Asymptotic Safety (ASG) [12].

4.1 Non-Commutative Geometry (The Hard Drive)

NCG postulates that spacetime has a minimum "pixel size" governed by the coordinate commutator:

$$[x^\mu, x^\nu] = i\theta^{\mu\nu} \quad (9)$$

where $\theta^{\mu\nu}$ is an antisymmetric tensor with dimensions of length squared. In canonical NCG models utilizing standard spectral triples [11], the coordinate non-commutativity restricts spatial localization, setting a minimum physical volume floor, $V_{\min} \approx \ell_P^3$. Consequently, as the universe rewinds to $t \rightarrow 0$, the volume cannot shrink below $V_{\min}$, meaning the Information Density ($\rho_\Phi = \Phi \cdot k_B T_{Pl} \ln 2/V$) hits a strict mathematical maximum.

4.2 Asymptotic Safety (The Processor Limit)

Simultaneously, Asymptotic Safety reveals that gravity's strength is scale-dependent. In the ultraviolet limit ($k \rightarrow \infty$), the dimensionless gravitational coupling $g(k) = G(k)k^2$ approaches a non-Gaussian fixed point g_* . In terms of the dimensionful Newton's constant, this implies $G(k) \sim g_*/k^2 \rightarrow 0$. At the Planck scale, gravity effectively weakens, acting as a regulator that halts the runaway gravitational collapse of information.

4.3 The Synthesized Origin Equation

Fusing these quantum gravity corrections with our Information Friedmann equation yields the master equation for the origin of the universe:

$$H^2(k) = \frac{8\pi G(k)}{3} \left(\frac{\Phi_{\max} k_B T_{Pl} \ln 2}{V_{\min}} \right) \quad (10)$$

As $k \rightarrow \infty$, $G(k) \propto k^{-2} \rightarrow 0$ and $V \rightarrow V_{\min}$. In the strict infinite-momentum limit, the product $G(k) \cdot \rho_\Phi \rightarrow 0$, yielding a vanishing expansion rate $H(k) \rightarrow 0$. Far from being an inconsistency, this vanishing limit represents a physical resolution of the initial singularity: the universe asymptotically approaches a static, stable informational boundary state in the deep ultraviolet, resembling the emergent universe scenario of Ellis and Maartens [15]. A dynamical cosmological expansion is subsequently initiated as the renormalization group scale runs down to lower energies ($k < k_{Pl}$), where the gravitational coupling $G(k)$ runs to non-zero values, triggering the Big Bang.

5 Limitations and Future Work

This paper presents a speculative theoretical framework. Integrated Information (Φ) is utilized here as a formal mathematical variable representing information integration, not as a claim that the early universe possessed consciousness. Furthermore, the topological mapping of variables across disciplines identifies structural mathematical analogies; establishing direct physical equivalence between neural thermodynamics and cosmological entropy would require empirical bridging laws.

A key limitation is that the connection between Φ and the holographic principle remains at the level of dimensional analysis. A rigorous derivation would require showing that the information-theoretic Φ satisfies the Bekenstein bound $S \leq A/(4\ell_P^2)$ for all causal horizons in the emergent universe scenario.

Additionally, a Φ -dominated emergent cosmology predicts specific observational signatures. In the ultraviolet limit ($k \rightarrow \infty$), the running of $G(k)$ slows the initial expansion rate, which would modify the primordial tensor-to-scalar ratio r and introduce a running of the spectral index n_s in the cosmic microwave background (CMB) power spectrum. Future work must rigorously derive the dynamics of H^2 in Eq. (10) using the exact ASG beta-functions to test these predictions against standard inflation, building upon asymptotic safety cosmology frameworks [16].

6 Conclusion

We have proposed the Information Singularity hypothesis, suggesting that the origin of the universe is not an infinite thermodynamic singularity, but a state of maximum, finite information density. By coupling Integrated Information (Φ) to the Einstein Field Equations, we show that information density can naturally saturate the critical density at $t \rightarrow 0$ within our framework. We suggest that Non-Commutative Geometry and Asymptotic Safety may provide a mechanism for a Holographic Ceiling, capping the volume and weakening gravity at the Planck scale. Finally, the thermodynamic bridge between neural entropy production and cosmic web entropy suggests a universal computational substrate governed by open-system Lindblad dynamics. This framework posits that reality may be fundamentally an information-processing engine, bounded by quantum and thermodynamic laws across all scales—a hypothesis that invites further theoretical and observational investigation.

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